

Jeffrey Navin

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PROFESSIONAL SUMMARY

Two internships, four shipped projects, one consistent thread: solve the problem, then prove it works. I bring strong data structures/algorithms fundamentals and object-oriented design to full-stack engineering — Python, Java, C/C++, TypeScript — across web platforms, ML-powered apps, GenAI systems, and embedded devices.

EXPERIENCE

Information Systems Intern – Chennai Petroleum Corporation Limited (CPCL)

Jun 2026 – Jul 2026

- Solved an ambiguous, undefined engineering problem end to end, measured by delivery of a fully working, tested application within the internship window, by independently designing and building a full-stack Document Intelligence system from requirements through deployment.
- Designed an efficient retrieval algorithm, measured by relevant, accurate results across multi-user test scenarios, by applying algorithmic thinking to implement top-K semantic similarity search over vector embeddings (PostgreSQL/pgvector).
- Built backend REST APIs (FastAPI, Python) and a Next.js/TypeScript frontend, then validated correctness and safety, measured by passing multi-user security and integration test suites, by applying object-oriented design through a Clean Architecture and a structured pre-delivery test plan.

Machine Learning Trainee – SmartBridge

May 2025 – Jun 2025

- Improved model performance on a classification task, measured by validation accuracy across experiment iterations, by applying feature engineering, data preprocessing, and model tuning with Pandas, NumPy, and Scikit-learn.
- Produced a reproducible evaluation workflow for candidate models, measured by consistent train/validation comparisons, by structuring exploratory data analysis and experimentation around clear evaluation criteria.

PROJECTS

Credit Card Approval Prediction System – Personal Project

- Built and deployed a credit card approval prediction web app, measured by real-time approve/reject predictions from live user input, by training a Scikit-learn classification model on historical applicant data and serving it through a Flask backend with an HTML/Bootstrap frontend.

Root – LLM-Powered Virtual Assistant – Personal Project

- Built Root, a desktop virtual assistant that dispatches natural-language commands to 21 modular tool handlers, measured by reliable intent routing across varied user phrasing, by designing a hybrid rule-based/LLM function-calling architecture and iterating on prompt design for intent classification.

CodeVault – Full-Stack Code Snippet Platform – Personal Project

- Shipped a full-stack CRUD platform with search, measured by secure, working JWT-authenticated APIs, by building a Node.js/Express backend, PostgreSQL schema, and React frontend.

SKILLS

Core CS: Data Structures & Algorithms, Object-Oriented Design, Complexity Analysis, Operating Systems

Programming: Python, Java, C, C++, SQL, TypeScript/JavaScript

Databases & Web: PostgreSQL, pgvector, REST APIs, React, Next.js, Node.js/Express, FastAPI

Practices: Unit & Integration Testing, Clean/Service-Oriented Architecture, Full SDLC, Version Control (Git/GitHub)

Generative AI: RAG pipelines, Prompt Engineering, LLM Function-Calling, Model Output Evaluation

ML Foundations: Pandas, NumPy, Scikit-learn, Feature Engineering, Model Evaluation

Tools: Git, GitHub, VS Code, Docker, Google Cloud (GCP)

CERTIFICATIONS

Machine Learning Internship (SmartBridge & Google) · Develop GenAI Apps with Gemini and Streamlit (Google Cloud) · Multimodal RAG with Gemini (Google Cloud)

EDUCATION

B.Tech Computer Science Engineering (AI & Robotics)

Aug 2023 – May 2027

Vellore Institute of Technology, Chennai